



Darshan
UNIVERSITY

Python Programming - 2301CS404

Lab - 14

Roll No.: 514

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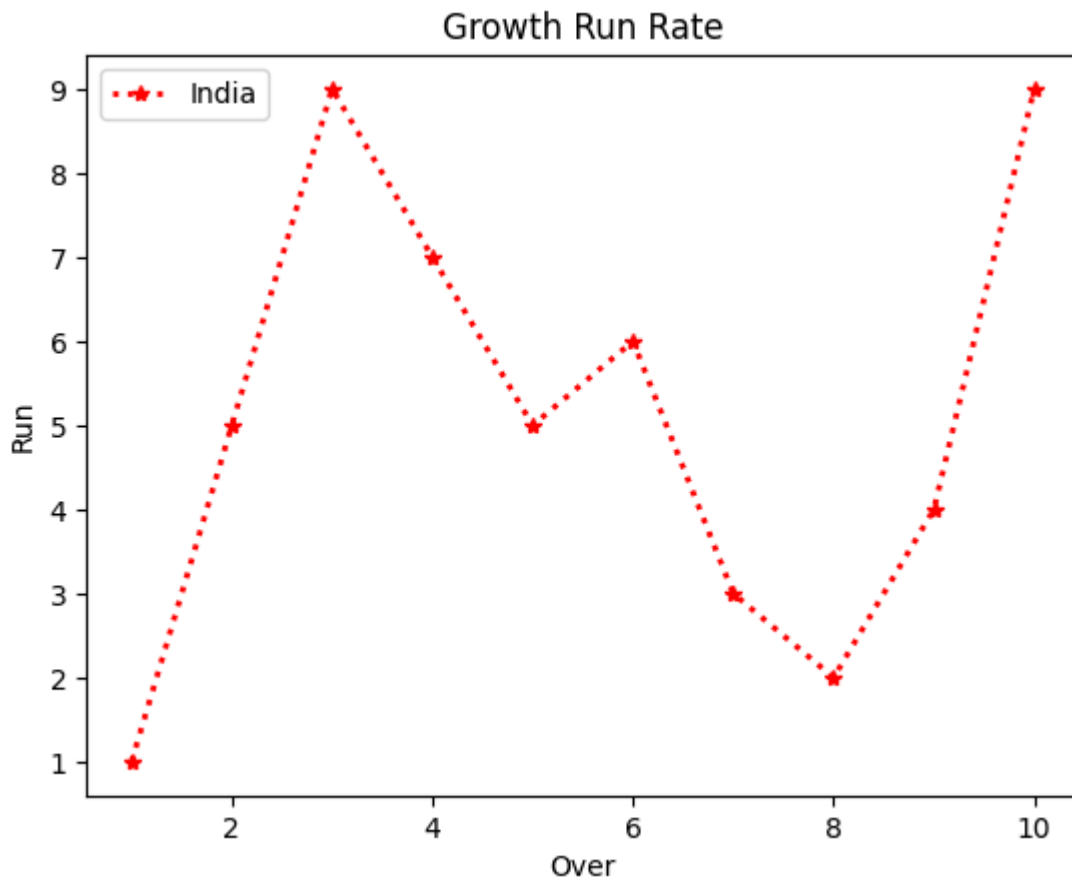
Import matplotlib below

```
In [11]: import matplotlib.pyplot as plt
```

01) WAP to display the line chart for given data for x and y axis.

```
In [13]: x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

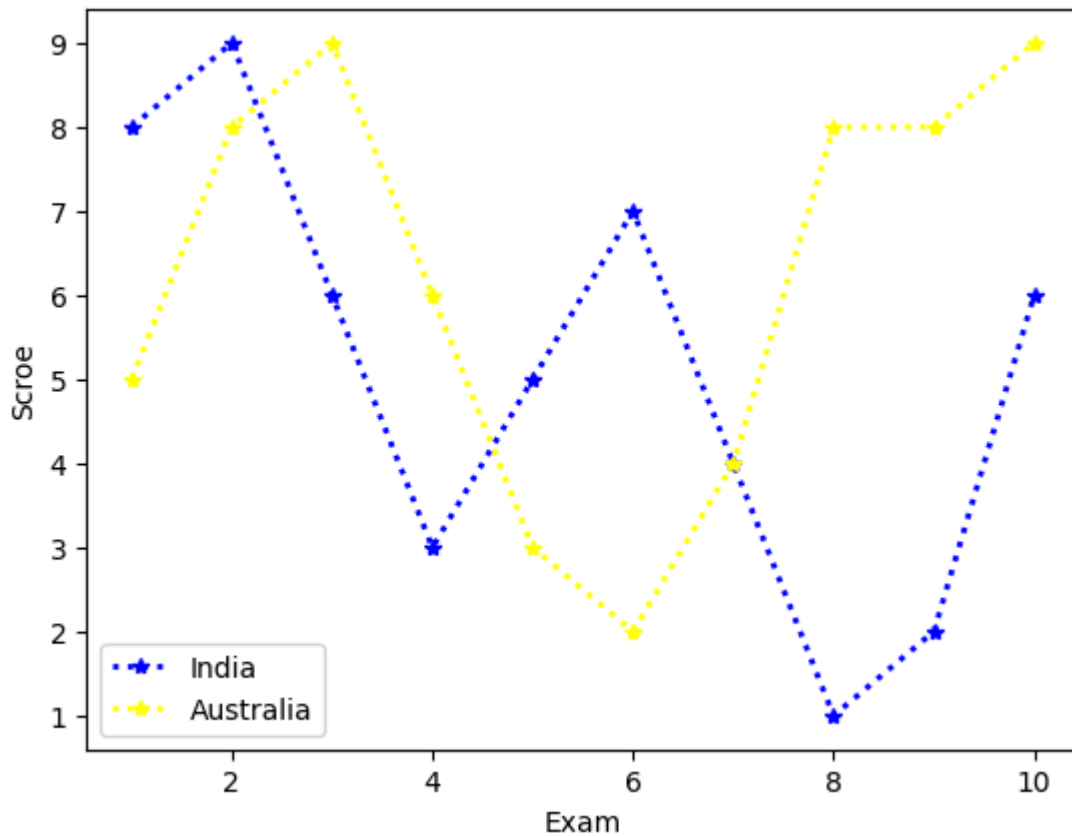
```
In [20]: plt.plot(x,y ,marker="*" , color="red" , linewidth="2" , linestyle=":", label="I")
plt.xlabel("Over")
plt.ylabel("Run")
plt.title("Growth Run Rate")
plt.legend()
plt.show()
```



02) WAP to generate line chart for given data.

```
In [22]: x = range(1,11,1)
cxMarks= [8,9,6,3,5,7,4,1,2,6]
cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

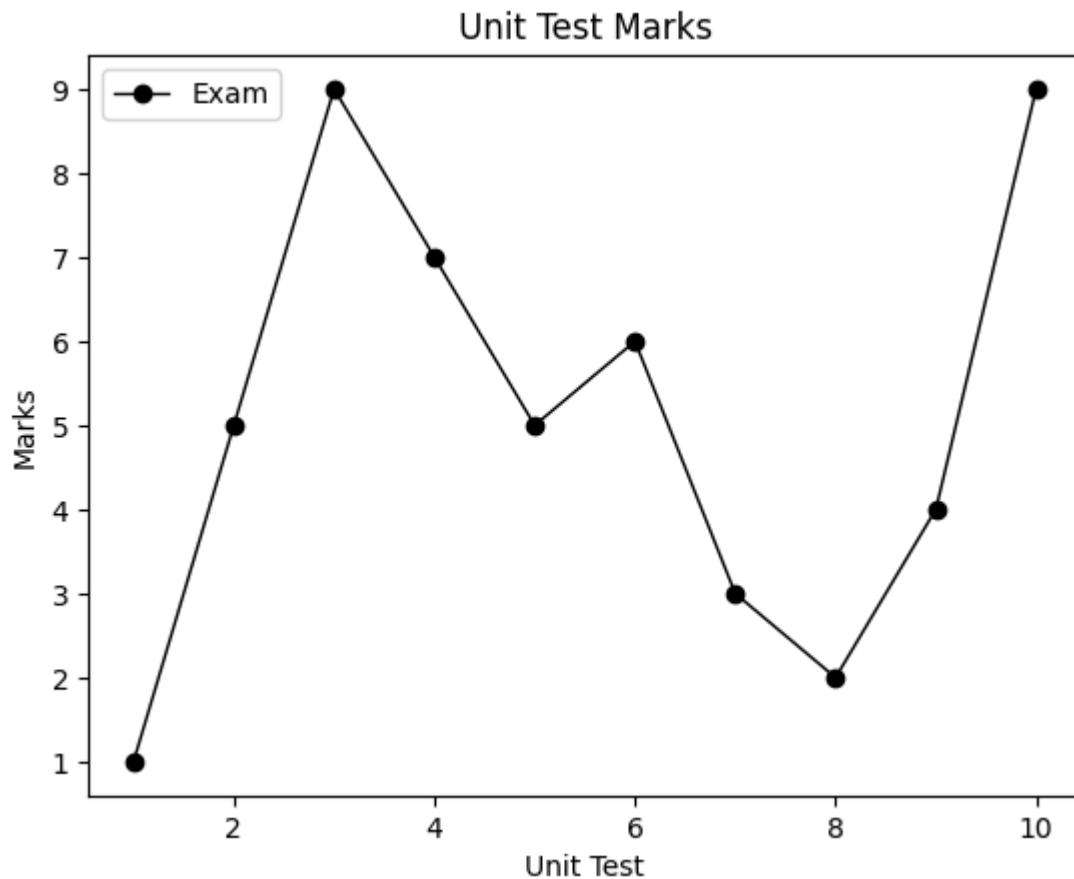
```
In [36]: plt.plot(x, cxMarks ,marker="*", color="blue" , linewidth = "2" , linestyle=":")
plt.plot(x , cyMarks , marker="*" , color="yellow" , linewidth = "2" , linestyle=":")
plt.xlabel("Exam")
plt.ylabel("Scroe")
plt.legend()
plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

```
In [37]: x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

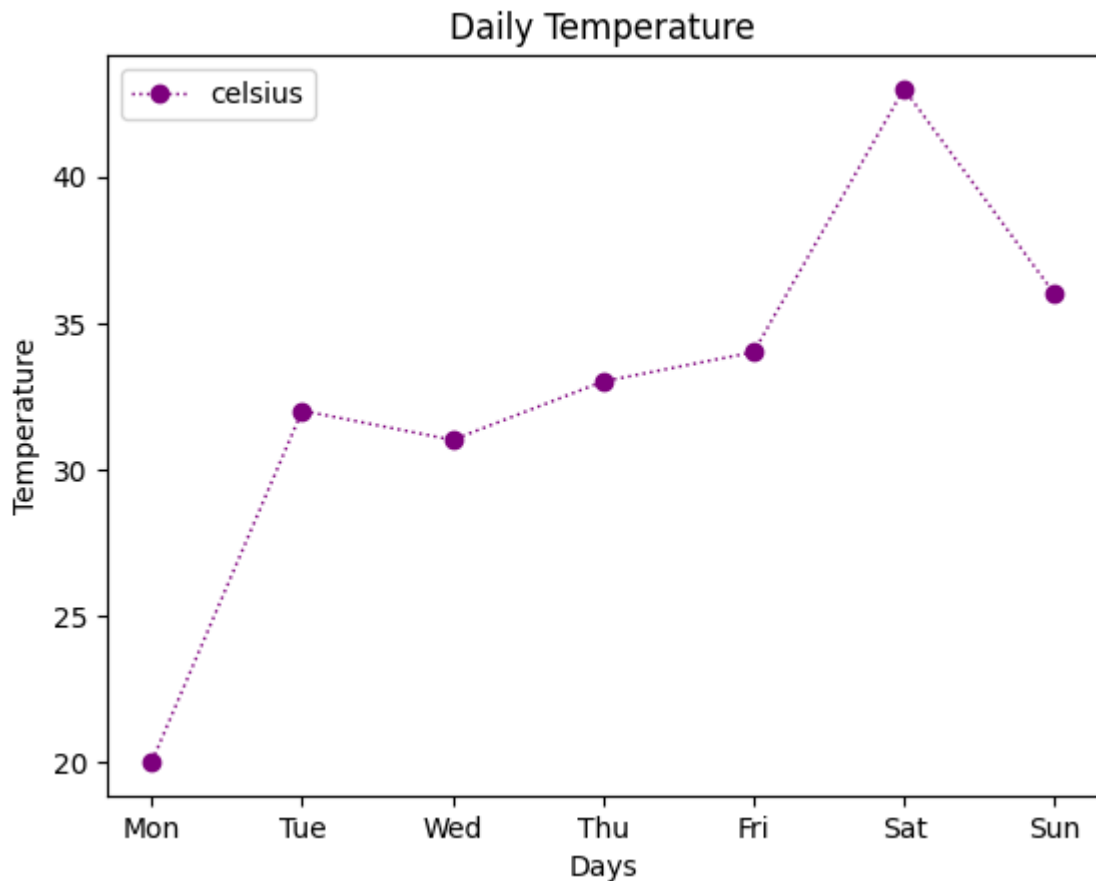
```
In [46]: plt.plot(x,y , marker="o" , color="black" , linewidth = "1" , label="Exam")
plt.xlabel("Unit Test")
plt.ylabel("Marks")
plt.legend()
plt.title("Unit Test Marks")
plt.show()
```



04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperature.

```
In [45]: Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]  
Temperature = [20, 32, 31, 33, 34, 43, 36]
```

```
In [48]: plt.plot(Days, Temperature, marker="o", color="purple", linewidth="1", line  
plt.xlabel("Days")  
plt.ylabel("Temperature")  
plt.legend()  
plt.title("Daily Temperature")  
plt.show()
```

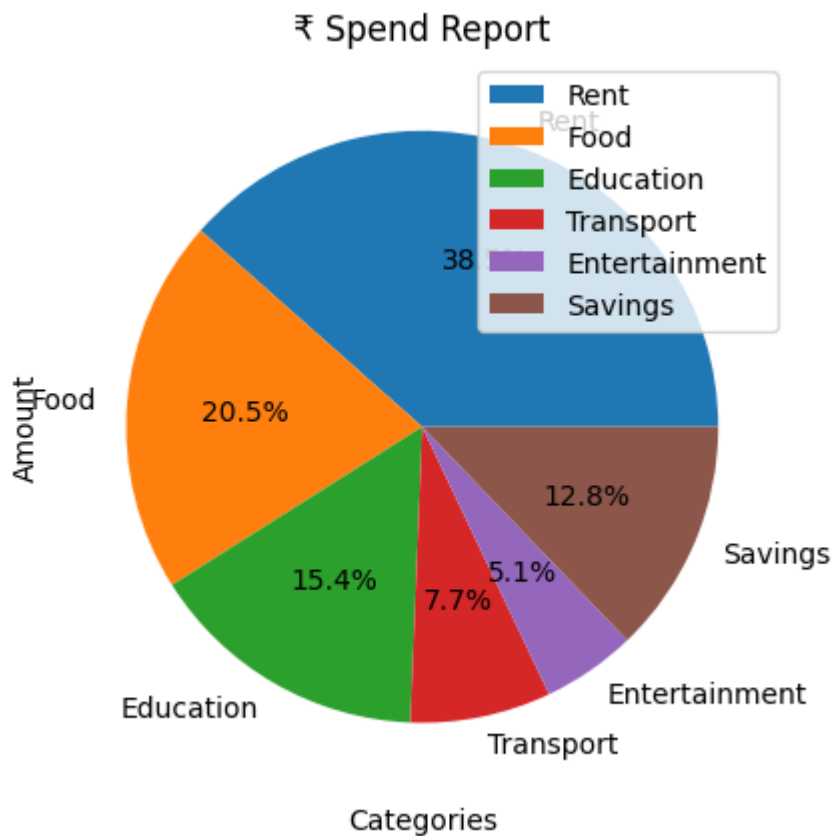


05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

```
In [61]: Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Saving"]
Amount = [15000, 8000, 6000, 3000, 2000, 5000]
```

```
In [107... plt.pie(Amount, labels=Categories, autopct="%1.1f%%")
plt.xlabel("Categories")
plt.ylabel("Amount")
plt.legend()
plt.title("₹ Spend Report")
plt.show()
```

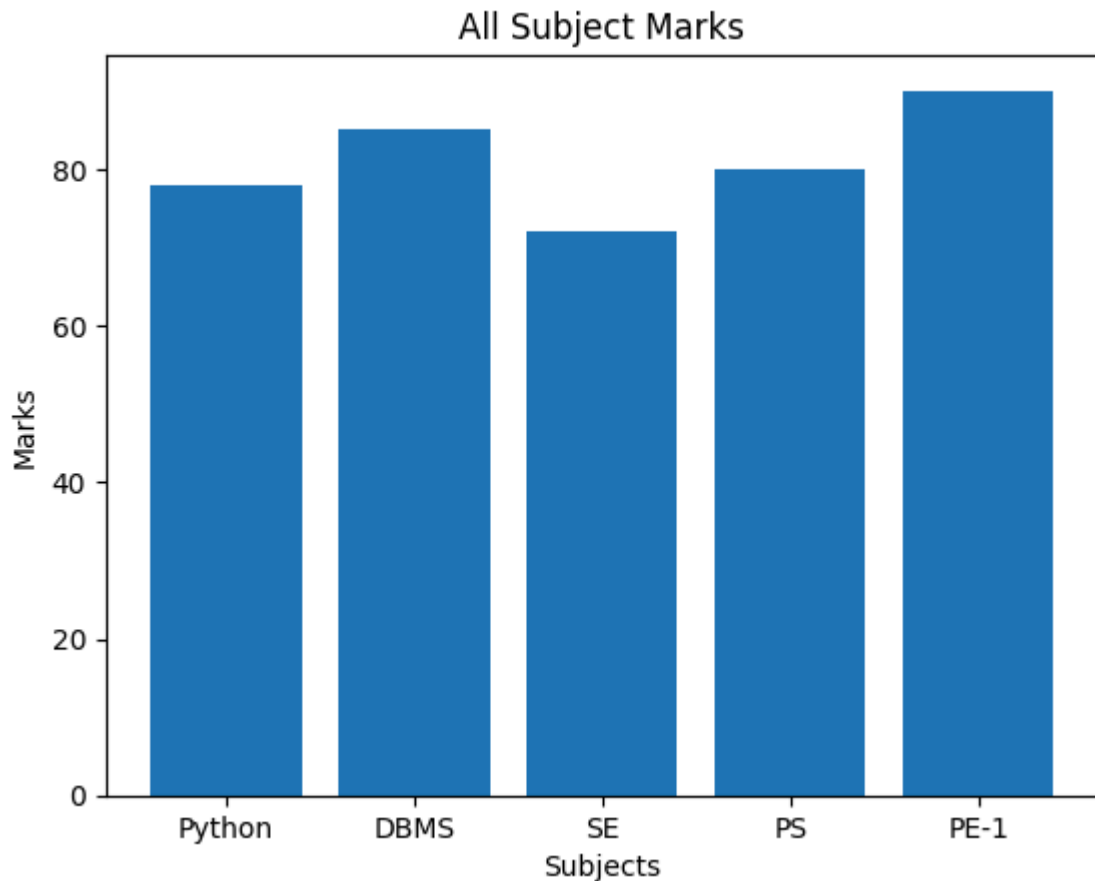


06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

```
In [70]: Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
Marks = [78, 85, 72, 80, 90]
```

```
In [104... plt.bar(Subjects,Marks )
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("All Subject Marks")
plt.show()
```

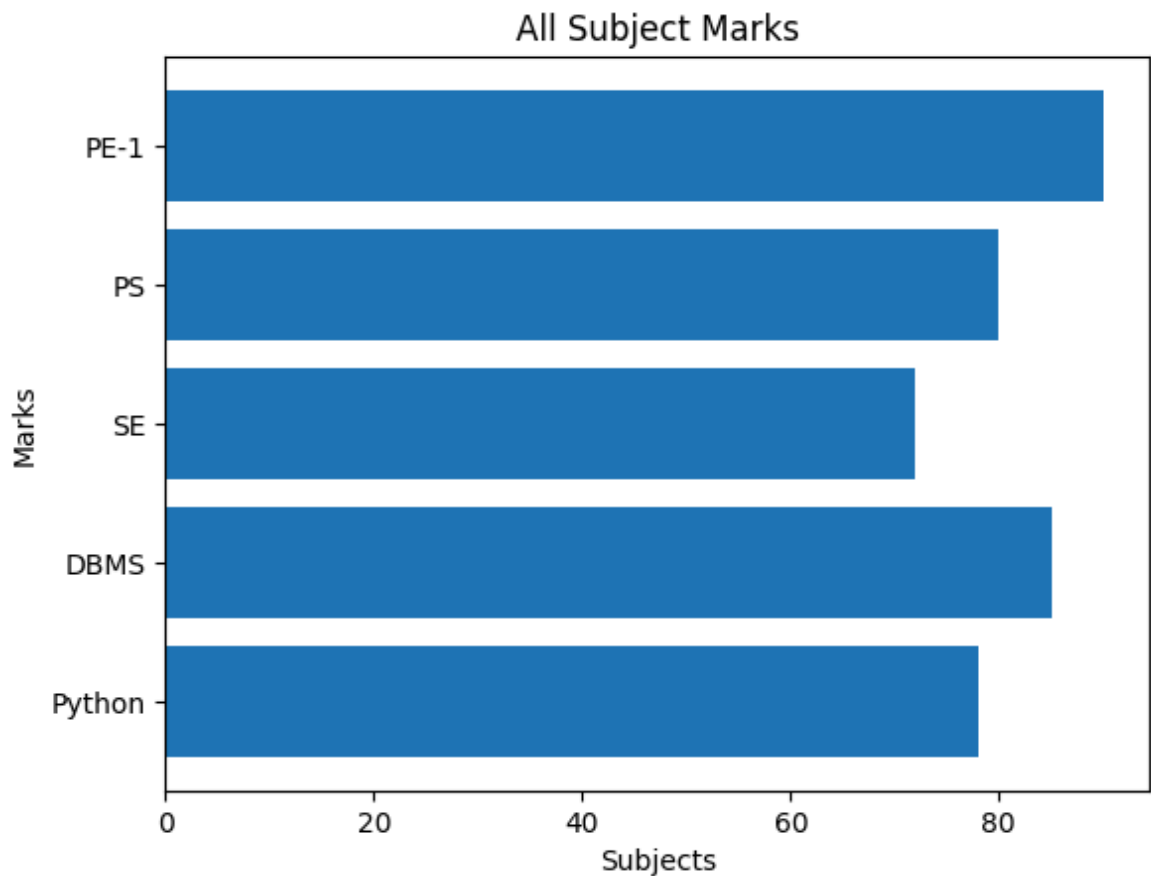


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

```
In [ ]: Categories = ["Fiction", "Science", "History", "Technology", "Biography"]
Books Sold = [120, 90, 70, 110, 60]
```

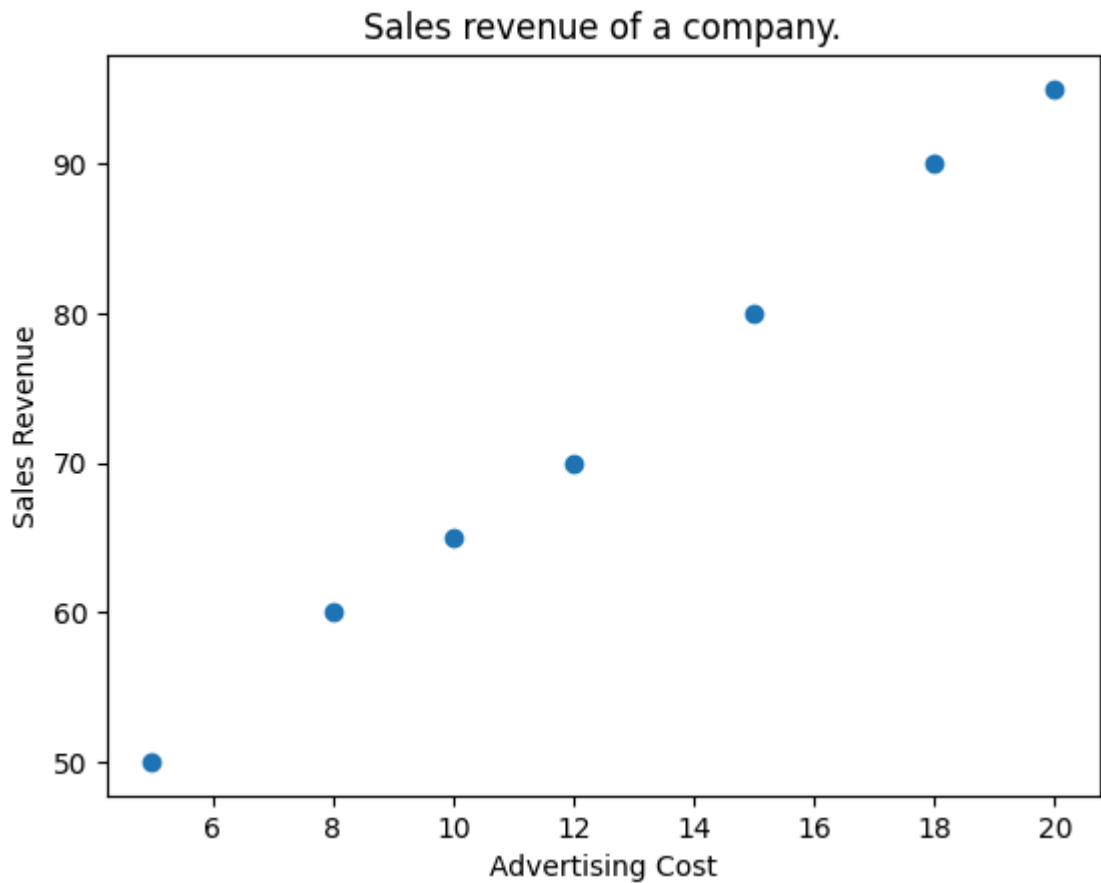
```
In [76]: plt.barh(Subjects,Marks)
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("All Subject Marks")
plt.show()
```



08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.

```
In [79]: Advertising_Cost= [5, 8, 10, 12, 15, 18, 20]
Sales_Revenue (₹ thousands) = [50, 60, 65, 70, 80, 90, 95]
```

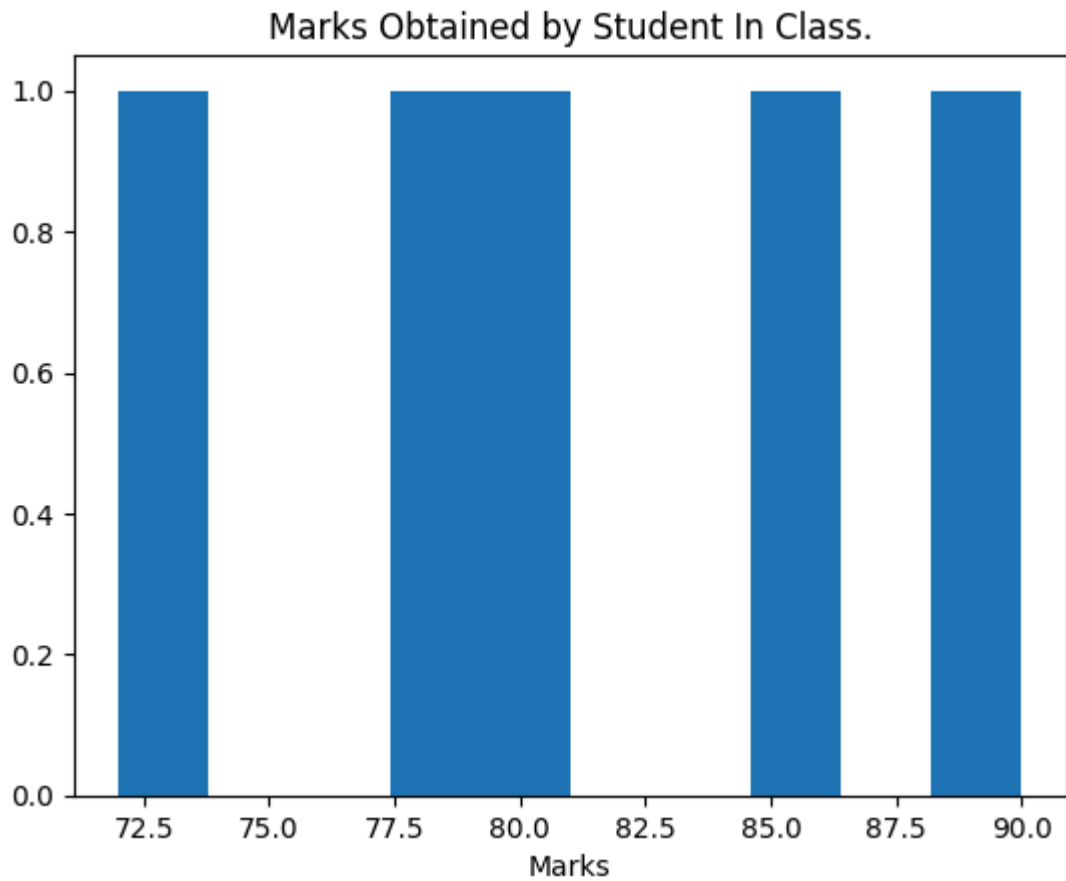
```
In [80]: plt.scatter(Advertising_Cost ,Sales_Revenue )
plt.xlabel("Advertising Cost")
plt.ylabel("Sales Revenue")
plt.title("Sales revenue of a company.")
plt.show()
```



09) WAP to create a histogram representing the distribution of marks obtained by students in a class.

```
In [ ]: Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

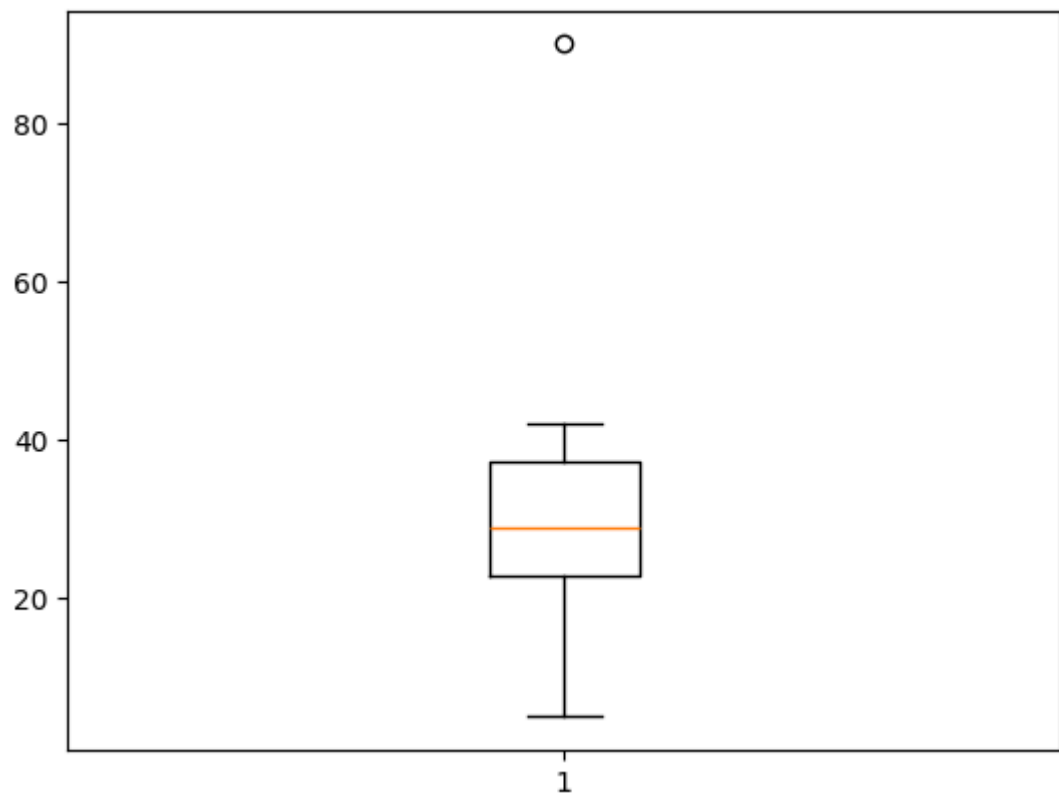
```
In [83]: plt.hist(Marks)
plt.xlabel("Marks")
plt.title("Marks Obtained by Student In Class.")
plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

```
In [98]: Commuting_Time= [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 90]
```

```
In [100... plt.boxplot(Commuting_Time)
plt.show()
```



In []: